Kollepara Naga Sai Manoj Abhiram

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Summary

Software Engineer with production experience in backend development and Generative AI. Proficient in Go and Python, with hands-on expertise building microservices, REST APIs, and Kafka-driven event-driven systems across CRM and EdTech platforms. Experienced in OCR pipelines, RAG workflows, Agentic AI, and domain-specific LLM fine-tuning. Passionate about building scalable, production-grade systems at the intersection of software engineering and AI.

Education

Shiv Nadar University Chennai — Chennai, Tamil Nadu
B.Tech Computer Science and Engineering (Internet of Things) — Sept 2021 – June 2025

Experience

Deepta AI — Hyderabad, Telangana
Software Engineer 1 — Aug 2025 – Present

- Designed OCR document-processing pipelines for CRM and EdTech platforms, turning unstructured inputs into structured data and cutting manual validation overhead by ~60%.
- Built RAG pipelines and Agentic AI workflows for CRM use cases, integrating LLMs and fine-tuning models on domain-specific data.
- Engineered Kafka-driven event pipelines in Go to stream real-time data across microservices, enabling high-throughput email and WhatsApp notifications for 10K+ users.
- Configured Omnichannel engagement platforms (WebEngage) to orchestrate email and WhatsApp campaigns, improving reach across CRM and EdTech products.
- Strengthened authentication reliability by building OTP and user verification flows with Redis-backed session management, cutting authentication latency by ~40%.
- Shipped 15+ REST APIs and CRUD modules in Go and PostgreSQL, powering listing pages, user interactions, and data management across CRM and EdTech platforms.

Projects

Pest Detection Application | *Deep Learning, Flask, React, AWS EC2, Selenium, Python*

- Scraped and labeled a custom pest image dataset of 5K+ images across classes from BugGuide using Selenium, training an Inception-based CNN with augmentation to address severe class imbalance.
- Deployed a Flask and React app on AWS EC2 achieving ~75% classification accuracy with real-time pest identification from uploaded images.

Automated Personal Finance Manager | *MERN Stack, Python, Machine Learning, NLP*

- Built a full-stack finance platform with ML-based transaction categorization using Random Forest achieving ~85% accuracy, and an NLP chatbot for financial Q&A.
- Delivered 6+ features including expense forecasting, interactive dashboards, budget alerts, multi-currency support, and a risk-profile-based stock recommendation module.

Technical Skills

Programming Languages: Go, Python, SQL, JavaScript
Backend and Tools: Kafka, Microservices, REST APIs, Flask, FastAPI, Streamlit, React, Selenium, Git, Postman
Databases and Caching: PostgreSQL, Redis, MongoDB
AI / GenAI: OCR, OpenCV, NLP, LLM Fine-Tuning, Prompt Engineering, RAG, Agentic AI, Vector Databases

Certifications

IBM Generative AI Engineering Professional Certificate (Coursera) | Building Microservices API in Go (Coursera) | Joy of Computing using Python (NPTEL) | Data Mining (NPTEL)